

Diversification leads to mediocrity... or does it?

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Diversification is one of the most often repeated principles in the world of investing. It is taught at universities, recommended by financial institutions and usually presented as an almost unquestionable rule. Taken to the extreme, however, it can become a costly exercise that ends up delivering mediocre results.

Imagine a portfolio that includes practically every available financial asset: stocks from various countries, government and corporate bonds, commodities, currencies, hedge funds, long-only and long/short strategies, and other asset classes. Suppose, too, that this portfolio holds some 200 different positions, optimized through a complex correlation matrix and rebalanced quarterly.

The result is an enormous amount of analysis, monitoring and administration. The reward? A 5% annual return with low volatility.

Now compare that effort with a much simpler alternative: buying a short-term, high-grade corporate bond that offers practically the same return and volatility. In this case, the management work is minimal.

The comparison seems to lead to a provocative conclusion: is diversification really worth it?

The answer is neither yes nor no. It depends.

The example above is deliberately extreme, but it illustrates a real phenomenon: excessive diversification can dilute the opportunities to create value

so much that it ends up as a strategy that barely replicates the market average. In other words, the more you try to eliminate specific risk, the harder it becomes to achieve significantly better results.

It would be a mistake, however, to conclude that diversification is useless.

Its main purpose was never to increase returns. Its real goal is to reduce concentration risk.

An investor who puts all of their capital into a single bond may obtain exactly the same expected return as a broadly diversified portfolio. But they are also fully exposed to a single adverse event: the issuer's default. Cases such as Uruguay's Conexión Ganadera are a reminder that even investments considered reasonably safe can end in heavy losses when the whole bet depends on a single asset.

Diversification does not eliminate risk; it simply prevents a single mistake, a fraud or a default from destroying the investor's wealth.

That is why it is important to separate two concepts that are often confused:

- Returns depend on the quality of the investments chosen and on the price paid for them.
- Diversification determines how much a negative event will affect total wealth.

They are different dimensions. One does not guarantee the other.

An overly concentrated portfolio can deliver extraordinary returns... or suffer catastrophic losses. At the opposite extreme, an excessively diversified portfolio will probably survive almost any event, but it will rarely stand out for its results.

The key is to find a balance: diversify enough that no single asset can jeopardize your wealth, but not so much that the portfolio becomes a mere collection of average investments.

Diversification is a risk-management tool, not a machine for generating returns. Anyone who expects better returns simply from adding more assets to a portfolio is chasing the wrong goal. The real question is not how many assets a portfolio holds, but whether each one adds value and earns its place in the investment strategy.

A case study

The Standard & Poor's 500 index (S&P 500) is generally considered the best reflection of the U.S. stock market. It is made up of the 500 largest companies listed in the United States, but there is a detail that often goes unnoticed: not all companies carry the same weight.

The index is weighted by market capitalization. This means that the greater a company's market value, the more influence it has on the index's performance. If one company is worth five times as much as another, it will carry roughly five times the weight.

For example, a company like NVIDIA, with a market capitalization of around \$5 trillion, has far more influence than companies like ExxonMobil, whose market value is around \$650 billion.

As a result, the 15 or so largest companies account for nearly 45% of the S&P 500, and most of them are technology companies or businesses closely tied to the tech and artificial intelligence ecosystem. In other words, although the index contains 500 companies, a very large part of its

performance depends on a small group of tech giants.

This leads to an interesting observation: by construction, the S&P 500 is already fairly concentrated.

The experiment

To analyze the effect of diversification, we built ten different portfolios using data from 2010 onward.

At the end of each September, the S&P 500 companies are ranked by market capitalization and the following strategies are formed:

- TOP1: only the largest company by market capitalization.
- TOP2: the two largest companies, weighted by market capitalization.
- TOP3: the three largest, weighted by market capitalization.
- ...
- TOP10: the ten largest companies, weighted by market capitalization.

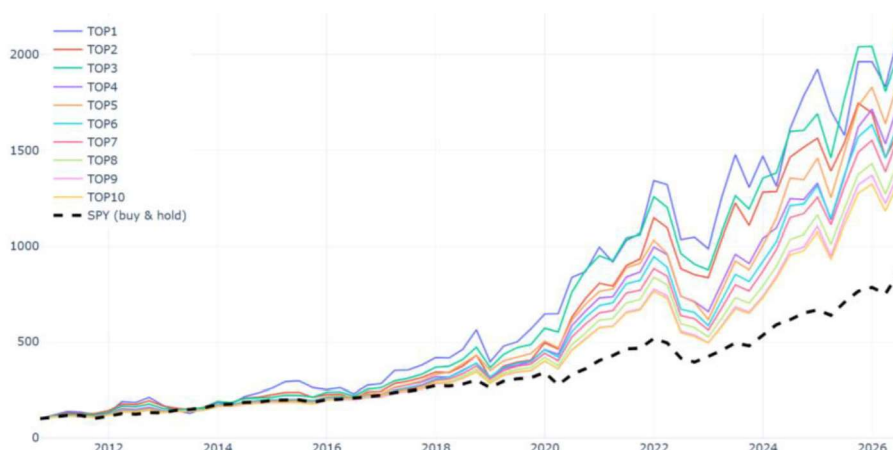
Each portfolio is held for one year and then rebuilt using the new market-cap ranking of the following September. Finally, all strategies are compared with SPY, the ETF that tracks the S&P 500.

The results are surprising.

Strategy	CAGR (%)
TOP1	21.35%
TOP2	19.24%
TOP3	20.98%
TOP4	19.95%
TOP5	20.46%
TOP6	19.49%
TOP7	19.15%
TOP8	18.49%
TOP9	18.16%
TOP10	17.90%
SPY	14.68%

(Annual rebalancing at the end of each September. CAGR (%) = Compound Annual Growth Rate)

ANNUAL rebalancing since 2010, Q3 anchor (September) — Oct-Sep holding (linear scale)
 TOP1..TOP10 by market cap vs SPY — base 100 = Sep-2010, total return



What do these results tell us?

At first glance, it would seem that concentrating investment in the largest companies produces returns considerably higher than those of the index itself.

Investing USD 100,000 in 2010 and buying only the top stock (TOP1) each September produced USD 2,105,949 in almost 16 years, compared with “only” USD 864,384 for SPY.

As more companies are added, returns tend to decline and the portfolio starts to look more and more like the S&P 500.

Does this mean that diversification destroys returns?

What these data show is that, during the period analyzed (2010 onward), the extraordinary appreciation of a small group of mega-caps—mainly technology companies—was so strong that concentrating investment in them proved more profitable than holding the entire index.

However, it is important to distinguish between a historical observation and an investment rule.

Concentration increased the returns obtained in that period, but it also increased the risk. Had any of those few companies suffered a serious setback, the impact on the portfolio would have been far greater than on the index.

Diversification does not seek to maximize expected returns. Its role is to reduce the specific risk of each company.

This experiment illustrates exactly that trade-off: the less you diversify, the greater the potential return... but also the greater the exposure to risk. The more you diversify, the closer returns move to the market's and the lower the volatility.

In short, diversification does not create returns. What it does is buy peace of mind, at the cost of part of the potential for extraordinary results. ■